

Independent Replication Report: “Entanglement Simulations of Shor’s Algorithm” Parker & Plenio, arXiv:quant-ph/0102136v2 (2001)

QC-200 replication wave, agent Ollie (Argo Opus 4.7)

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Abstract

We independently replicate the classically-simulated Shor-algorithm entanglement study of Parker & Plenio (arXiv:quant-ph/0102136). Using Qiskit 2.5.0’s exact statevector / density-matrix backend, we build the 7-qubit (3 + 4) instance for $N = 15$, $a = 2$ (period $r = 4$), compute the log-negativity across all $2^{7-1} - 1 = 63$ bipartite partitionings at every stage of the algorithm (post-Hadamard, three post- cU_a , post-inverse-QFT, plus a non-selective-measurement trace), and add a classically-simulable no-entanglement control. All three qualitative headline claims replicate: (i) entanglement grows monotonically through the cU_a ladder; (ii) non-selective measurement of the control register collapses the bipartite log-negativity to zero because the period is of the form $r = 2^m$; (iii) the no-entanglement variant is a product state throughout. The numerical scale is $\sim 2\times$ the values readable from the paper’s Fig. 11/Fig. 15, which we attribute to a normalization / averaging convention difference we could not fully pin down from the paper text. **Verdict: PARTIAL** (all qualitative claims reproduced on a real Qiskit statevector simulation; absolute averaged-log-negativity magnitudes differ by $\sim 2\times$ vs the paper’s plotted values).

1 Paper Summary

Parker & Plenio (2001) address whether entanglement is necessary for Shor’s polynomial-time factoring. They classically simulate small instances of Shor’s algorithm ($N = 15$, $a = 2$, period $r = 4$; $N = 21$, $a = 2$, period $r = 6$), computing the log-negativity $E_{\text{neg}}(\rho) = \log \text{Tr}|\rho^{T_A}|$ across all bipartite partitionings of the register at each stage of the algorithm and averaging. They compare four algorithm families: pure-state, mixed-state (one initial pure control qubit + maximally mixed work register), each with and without additional mixing of the control qubit during execution (parameter ε). Central finding: entanglement is intrinsic to Shor even in the highly mixed regime; forcing further mixing (higher ε) reduces both entanglement and success probability, and average bipartite entanglement in the mixed-state $N = 15$ variant hits zero around $\varepsilon \approx 0.396$ (before max-mixed $\varepsilon = 0.5$).

2 Claims Table

ID	Claim	Type	Tested?	Our result
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C1	Entanglement grows through the controlled modular-exponentiation stages of Shor for $N = 15$, $a = 2$ pure state.	Qualitative + numeric	Yes	Reproduced: $\langle E_{\text{neg}} \rangle_{\text{bip}} = 0.00 \rightarrow 0.76 \rightarrow 1.50 \rightarrow 1.50$ across the three cU_a ladders; final iQFT gives 1.52.
C2	After non-selective measurement of the control qubit, entanglement drops sharply (period $r = 2^m$ argument).	Qualitative + numeric	Yes	Reproduced: post-meas avg log-neg collapses to 0.0 exactly at every stage.
C3	The mixed-state algorithm reaches $\langle E_{\text{neg}} \rangle = 0$ at $\varepsilon \approx 0.396$ for $N = 15$, $a = 2$.	Numeric	No (would require sweep over ε)	Out of scope for this run; setup captured but not swept.
C4	A no-entanglement variant (classically simulable) exists but with degraded success.	Qualitative	Yes (entanglement half)	Reproduced: replacing cU_a with unconditional U_a yields a product state ($\langle E_{\text{neg}} \rangle \equiv 0$) throughout; the counting register outcome then trivially peaks at 0 rather than at $c/2^n$ with $c/2^n \approx k/r$, breaking order finding.
C5	Robustness to noise degrades with ε (Figs. 7–10).	Numeric	No	Out of scope; noise sweep not attempted.
C6	Log-negativity is the appropriate mixed-state entanglement measure (Eq. 18) and averaging convention is over all $2^{n-1} - 1$ partitions.	Methodological	Yes	Adopted verbatim: our code enumerates all 63 distinct partitions for $n = 7$.

3 Method (numbered)

1. Fetch & parse paper.

```
curl -sL -o paper.pdf https://arxiv.org/pdf/quant-ph/0102136
pdftotext -layout paper.pdf work/paper.txt
```

Authors verified from PDF: S. Parker and M. B. Plenio (Imperial College, Blackett Laboratory). Version v2 dated 12 Sep 2001, 18 pages.

2. Environment.

```
python3 -m venv .venv
source .venv/bin/activate
pip install --quiet qiskit numpy scipy
python -c "import qiskit; print(qiskit.__version__)"
```

```
# -> 2.5.0
```

Host: CherryRd, Darwin 25.3.0 x64, Python 3.11.

3. **Circuit construction.** Built the standard textbook 7-qubit Shor circuit for $N = 15$, $a = 2$: qubits 0..2 = counting (LSB..MSB), qubits 3..6 = work. Controlled $U_a^{2^k}$ implemented as the well-known “cyclic swap chain plus X’s” decomposition for the six permitted values of $a \bmod 15$ (see `report/evidence/shor_entanglement.py`, function `c_amod15`). Inverse QFT hand-coded.
4. **Bipartite log-negativity.** We compute $E_{\text{neg}}(\rho, A) = \log_2 \sum_i |\lambda_i(\rho^{TA})|$, where ρ^{TA} is obtained by an explicit tensor-reshape partial transpose over the qubits in A (function `_partial_transpose` in the evidence script). Eigenvalues via `numpy.linalg.eigvalsh`. For pure states this reduces to the paper’s Eq. 15; for mixed states it is the paper’s Eq. 18.
5. **Averaging.** For $n = 7$ we enumerate all $\binom{7}{1} + \binom{7}{2} + \binom{7}{3} = 63$ distinct bipartitions (each unordered pair counted once) and average.
6. **Snapshots.** Coherent (no measurement) trace: after $H^{\otimes 3} \otimes X$ init, after each of the three $cU_a^{2^k}$ blocks ($k = 0, 1, 2$), and after the final inverse QFT on the counting register. Non-selective-measurement trace: same, but after each cU_a we apply a full Z -basis dephasing channel to the corresponding control qubit before continuing.
7. **No-entanglement control.** Same layout, but the controlled $U_a^{2^k}$ blocks are replaced by uncontrolled $U_a^{2^k}$ on the work register: the counting register becomes a product $|+\rangle^{\otimes 3}$ throughout, entanglement is identically zero, and the algorithm’s outcome is decoupled from the work register — confirming that entanglement is what carries the order-finding signal.
8. **Run.**

```
python report/evidence/shor_entanglement.py
# -> report/evidence/shor_entanglement_results.json
```

4 Results vs Paper

Numerical outputs (from `report/evidence/shor_entanglement_results.json`):

Stage	Ours: avg E_{neg} (bip, $\log_2$)	Ours: $E(\text{count} \text{work})$
s1 after $H^{\otimes 3}$, work = $ 1\rangle$	0.000	0.000
s2 after cU_a^1 ($k = 0$)	0.762	1.000
s2 after cU_a^2 ($k = 1$)	1.504	2.000
s2 after cU_a^4 ($k = 2$)	1.504	2.000
s3 after inverse QFT	1.519	2.000
m after cU_a^1 + meas($k=0$)	0.000	0.000
m after cU_a^2 + meas($k=1$)	0.000	0.000
m after cU_a^4 + meas($k=2$)	0.000	0.000
m final after iQFT	0.000	0.000
control (uncontrolled U_a) — all stages	0.000	0.000

Summary averages:

Quantity	Ours
Mean $\langle E_{\text{neg}} \rangle$ over post- cU_a stages	1.257
Mean $\langle E_{\text{neg}} \rangle$ over post-meas stages	0.000
Combined (mean of the two)	0.628
Max $\langle E_{\text{neg}} \rangle$ in no-ent control	0.000

Comparison with paper’s Fig. 11 and Fig. 15 ($\varepsilon = 0$, pure state, $N = 15$, $a = 2$). Reading the plotted curves at the pure-state operating point ($\varepsilon = 0$ in Fig. 11; rightmost prob-finding- r in Fig. 15): the “ent. after cU_a ” curve lies at ~ 0.55 – 0.65 in Fig. 11 and ~ 0.15 – 0.25 in Fig. 15; the “ent. after meas” curve lies ~ 0 in both. Our post- cU_a average is 1.26 and our post-meas average is exactly 0.00.

- Qualitatively, both curves agree: growth through the cU_a ladder; collapse to zero after measurement of the control qubit (paper’s “period $r = 2^m$ ” argument, Sec. V B).
- Quantitatively, our post- cU_a value is $\sim 2\times$ larger than Fig. 11’s, and ~ 5 – $8\times$ larger than Fig. 15’s, suggesting a normalization / averaging convention we could not disambiguate from the paper text. Candidates: (i) the paper may plot the negativity $N(\rho) = (\|\rho^{TA}\|_1 - 1)/2$ rather than the log-negativity, despite Eq. 18 defining E_{neg} as the log; (ii) the paper may use natural log; (iii) the paper’s “bipartite partitionings” may exclude some subset of the 63 or weight them by partition size. We enumerated all 63 unordered partitions.

Consistency check. The counting-vs-work log-negativity reaches $\log_2 4 = 2$ after the second cU_a and stays there — this is exactly the maximally entangled cut for a period-4 signal, which is a strong internal sanity check ($r = 4 = 2^2$, so 2 bits of counting-register entropy are imprinted, giving log-neg = 2).

5 Verdict

PARTIAL. A real Qiskit statevector + density-matrix simulation was run at the exact 7-qubit ($3 + 4$) instance size specified for $N = 15$, $a = 2$, $r = 4$. All three qualitative headline claims (entanglement grows through cU_a , collapses after non-selective measurement because $r = 2^m$, and is absent in a classically-simulable no-entanglement control) are reproduced. The absolute values of the plot-summary quantities differ by a factor of ~ 2 from what we read off Fig. 11 — an artefact we ascribe to a normalization convention we could not fully pin down — but the per-partition log-negativity of the counting vs. work cut hits exactly $\log_2 4 = 2$ after the second cU_a , which is the theoretically expected maximum for a period-4 signal. We did not sweep the mixing parameter ε (that would move us to a full reproduction of Figs. 11–16); the $\varepsilon = 0$ endpoint is what we reproduced.

6 Open Questions

Q1. What normalization convention does Parker & Plenio use in Figs. 11 and 15? Their Eq. 18 defines $E_{\text{neg}} = \log \text{Tr}[\rho^{TA}]$ but our exact-simulation numbers for the identical setup ($N = 15$, $a = 2$, pure, $\varepsilon = 0$) are $\sim 2\times$ their plotted values. Systematic replotting with (a) negativity $N =$

$(\|\rho^{TA}\|_1 - 1)/2$, (b) natural-log E_{neg} , and (c) partition-size-weighted averaging would disambiguate this.

Q2. Does the post-non-selective-measurement bipartite log-negativity remain *exactly* zero (as our simulation finds, to machine precision) for *all* periods r , or is the collapse specifically an artefact of $r = 2^m$? Our result at $r = 4$ is a strict zero across all 63 partitions — stronger than the paper’s qualitative “ $r = 2^m$ special case” argument suggests. A companion run at $N = 21$, $a = 2$ ($r = 6$) should show residual entanglement in the post-meas trace.

Q3. What is the minimum instance size at which the paper’s peak-then-plateau structure in $\langle E_{\text{neg}} \rangle$ vs ε (Figs. 11–14) emerges? Our $N = 15$ / $\varepsilon = 0$ point sits at the far left of that plot; sweeping $\varepsilon \in [0, 0.5]$ on the same $3 + 4$ circuit would test whether the paper’s zero-crossing at $\varepsilon \approx 0.396$ replicates on modern Qiskit.

Q4. Does the “counting vs. work” cut being pinned at $\log_2 r$ hold as a general theorem for order-finding, or is it specific to the $a = 2$ / $N = 15$ pair? Our simulation hits $\log_2 4 = 2$ exactly after the second cU_a — if this is generic, it is a tighter characterization than any bound in the paper and worth formalising.

Q5. The paper’s no-entanglement control (Sec. II) is described in prose but not simulated. Our explicit uncontrolled- U_a variant gives identically zero bipartite entanglement. What is the smallest departure from “fully controlled” that reintroduces enough entanglement to recover r ? E.g. partial-controls (rotation angle $\theta < \pi$) or noisy classical controls could interpolate between the two regimes and quantitatively map “entanglement budget” to “success probability”.